



Kishoreganj University
3rd Year 1st Semester B. Sc. (Engg.) Final Examination-2024
Department of Computer Science and Engineering
CSE 3101: Data Communication (3 Credits)

Time: 3 Hours

Full Marks: 70

Figures shown in the right margin indicate full marks.

Answer 05 out of 07 questions.

- ✓1 a. What are the three criteria necessary for an effective and efficient network? Explain in details. 4
- b. What is the relationship between protocols and standards in networking? Describe the key elements of protocol with examples. 2.5+2.5=5
- c. Explain in detail the different types of transmission impairments that affect data communication over a network. 5
- ✓2 a. What is frequency domain representation? Why it is used in analyzing signals? 1.5x2=3
- b. State and explain Shannon channel capacity. Consider a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level? 6
- c. The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km? 5
- 3 a. List down the characteristics of different line coding schemes. 4
- b. Differentiate multiline and multi-level encoding techniques with suitable examples.
- c. State the Nyquist theorem. Illustrate the components of PCM encoder and briefly explain the steps of PCM. 1+3=4
- d. Draw the signal representation of data 01001110 in NRZ-L, NRZ-I, Polar RZ, Manchester, Differential Manchester and AMI encoding scheme. Assume the signal level at $t(0)$ is positive. 6
- [Hint: Draw the clock pulse of master clock and draw the signals with respect to the clock pulse.]
- ✓4 a. What is modulation? Why analog to analog conversion is needed in data communication? 3
- b. Which of the four digital-to-analog conversion techniques (ASK, FSK, PSK or QAM) is the most susceptible to noise? Defend your answer. 3
- c. Find the bandwidth for the following situations if we need to modulate a 5-KHz voice. 3
- i. AM
- ii. PM (set $\beta = 5$)
- iii. PM (set $\beta = 1$)
- d. Why PM is preferred over AM and FM? Describe how PM can be implemented with proper diagram. 5
- ✓5 a. What strategies are followed when the data rates of the input sources in a time-division multiplexing (TDM) system are not the same? Explain. 4
- b. We have four sources, each creating 250 characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find: 5
- (i) the duration of each character in each source,
- (ii) the frame rate,
- (iii) the duration of each frame,
- (iv) the number of bits in each frame, and
- (v) the data rate of the link.
- c. Describe the concept of frequency hopping spread spectrum (FHSS). How it achieves antijamming facility in data communication? 5

- 6 a. For CRC, given data word is 1010011010 and the divisor is 1011. 04
- Using binary division, show the generation of the codeword at the sender site.
 - Using polynomials, show the checking of the codeword at the receiver site.
- b. Show the complete process of checksum if a sender needs to send four data items 0x3456, 0xABCC, 0x02BC and 0xEEEE. Find the followings: 05
- Checksum at the sender site
 - Checksum at the receiver site if second data item is changed to 0xABCE.
- c. Hamming code is used to detect and correct error. Using Hamming code technique, find the codeword for the dataword 1011001. 05
- 7 a. Why microwave cannot penetrate wall while radio wave can? 2
- b. Why optical fiber is the backbone is today's network? Briefly describe the different classes of optical fiber including single mode, multimode step index, and multimode graded index, explaining their core structure, light propagation, and common applications. 1+6=7
- c. What is the significance of the twisting in twisted-pair cable? 3
- d. How does sky propagation differ from line-of-sight propagation? 2